

Allamaprabhu Ani

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Research Statement

PhD student developing **PhAST**, a matrix-free, autograd-driven phase-field fracture solver in PyTorch with neural-operator constitutive laws. Research bridges computational mechanics and machine learning: differentiable physics for inverse problems, learned closures for multiscale fracture, and benchmark methodology for the field. Lead author of the 2026 arXiv preprint on the solver and of the 2026 *Engineering Fracture Mechanics* review (Vol. 332, Art. 111778) on machine learning for fracture mechanics with collaborators at EPFL, University of Florida, and City St George's.

Education

PhD, Mechanical Engineering Jul 2023 – Jun 2026 (expected)

City, St George's, University of London – School of Sciences and Technology

Thesis: *Matrix-free differentiable fracture mechanics in PyTorch: inverse analysis, GPU solvers, and learned constitutive models.*

Supervisor: Dr Sathiskumar A. Ponnusami FRAeS.

MTech, Design and Manufacturing

Sep 2021 – May 2023

National Institute of Technology Silchar – CGPA 9.88/10. Best Student Award.

Thesis: Machine learning for fatigue life prediction in aluminium alloys. Advisor: Dr A.B. Deoghare.

BE, Mechanical Engineering

Aug 2016 – Jul 2020

GM Institute of Technology, Davangere (Visvesvaraya Technological University) – CGPA 8.47/10.

Awards and Honours

- **Yeoman and Travelling Scholarship, Worshipful Company of Tin Plate Workers** (Mar 2026). One of three Travelling Scholarship recipients across the City of London, for “AI-accelerated modelling for fracture prediction.”
- **Dean's Award for Outstanding Teaching Support**, School of Sciences and Technology, City St George's, University of London (Apr 2026). Nominated by students.
- **Associate Fellow of the Higher Education Academy (AFHEA)**, Advance HE (Oct 2024). Credential PR301728.
- **Funded PhD Studentship** “Modelling for Failure Analysis,” City St George's, University of London (2023).
- **Best Student Award**, MTech Design and Manufacturing, National Institute of Technology Silchar (2023).
- **Inspiring Student Award for Mechanical Engineering**, Visvesvaraya Technological University (Dec 2020).

Publications

Preprints

- P1. **Ani, A.S.**, Molinari, J.-F., Subhash, G., Ponnusami, S.A. (2026). A matrix-free, differentiable PyTorch solver for phase-field fracture: Formulation, benchmarks, and inverse analysis. [arXiv:2606.23458](https://arxiv.org/abs/2606.23458).

Journal articles

- J1. **Ani, A.S.**, Nakka, R., Subhash, G., Molinari, J.-F., Ponnusami, S.A. (2026). Machine learning for computational fracture and damage mechanics: Status and perspectives. *Engineering Fracture Mechanics*, 332, 111778. [doi:10.1016/j.engfracmech.2025.111778](https://doi.org/10.1016/j.engfracmech.2025.111778). *Open access; field-level review paper.*
- J2. Srinivasa, C.V., **Ani, A.S.** (2020). Protective Coatings for Bio-Composites: A Review. *IOP Conference Series: Materials Science and Engineering*, 925, 012048. [doi:10.1088/1757-899X/925/1/012048](https://doi.org/10.1088/1757-899X/925/1/012048).

Book chapters and conference papers

- C1. **Ani, A.S.**, Deoghare, A.B. (2024). Leveraging Machine Learning for Enhanced Fatigue Life Prediction in Aluminium Alloys. In: *Lecture Notes in Mechanical Engineering*, Springer. [doi:10.1007/978-981-97-7535-](https://doi.org/10.1007/978-981-97-7535-)

In preparation

- Two-part submission to *Computer Methods in Applied Mechanics and Engineering* (CMAME) on differentiable phase field fracture and autograd-based inverse problems. Drafts in revision; submission target Q3 2026.

Research Experience

PhD Student, Computational Mechanics and Machine Learning

Jul 2023 – present

City, St George's, University of London & Queen Mary University of London

- Designed and built **PhAST**, a matrix-free, PyTorch-native, GPU-accelerated phase-field fracture solver with autograd differentiation and neural operator hooks for learned constitutive laws.
- Demonstrated inverse analysis via gradient-based optimisation: scalar fracture-toughness recovery, spatial G_c -field reconstruction, and stiff inclusion localisation, all driven by autograd through a staggered solver.
- Cross-validated against Akantu (C++), FEniCSx, COMSOL, PhaseFieldX and PhaFiDyn on quasi static and dynamic benchmarks (Borden, Bourdin, Miehe, Kalthoff, SENT, Brazilian disc).
- Maintained reproducible HPC workflow on Hyperion2 (Slurm) with local Mac CPU and WSL CUDA development environments.

Visiting Researcher, LSMS Lab, EPFL

2025

Host: *Prof. Jean-François Molinari* (École Polytechnique Fédérale de Lausanne).

- Lead author of the *Engineering Fracture Mechanics* 2026 review of machine learning for computational fracture and damage mechanics.
- Engaged with the LSMS group on differentiable methods for solid mechanics; visit informed the autograd-through-staggered-solver framework in the PhD code base.

MTech Researcher, National Institute of Technology Silchar

2021 – 2023

Advisor: *Dr A.B. Deoghare*.

- Machine-learning surrogate models for fatigue life prediction across multiple aluminium alloy compositions (scikit-learn, MATLAB). Best Student Award.

Founder and Tool Developer, Aeroknacks India Ltd

2020 – 2021

- Built a library of validated aerospace hand-calculation automation tools and Excel-VBA workflows based on E.F. Bruhn, Michael Niu, Boeing Design Manuals, and ESDU references.
- Implemented Cozzone f_0 plastic-bending extraction, lug strength (Bruhn/Niu/ESDU), fastener lap-joint load transfer, column buckling (Euler/Johnson + AFFDL coefficients), modal analysis (Roark validation), ABD calculator, Neuber non-linear stress, and a NASTRAN .f06 parser. Tools shipped to an aerospace engineering firm.

Research Intern, NMCAD Lab, Indian Institute of Science Bengaluru**Research Intern, CGPL Lab, Indian Institute of Science Bengaluru** (Feb 2018) – Pressure Swing Adsorption for 99.997%-purity hydrogen from biomass gasifier.**Teaching and Outreach**

Teaching, City St George's (2023 – present)

- Co-pilot, Learning Enhancement and Development (LEaD) educational technology team (Sep 2024 – Feb 2025).
- Graduate Teaching Assistant, EG1004 Introduction to Mechanics, Materials and Manufacturing.
- Graduate Teaching Assistant, EG1002 Wind Turbine Challenge: Tower Design, and EG1002 Siemens NX CAD teaching support.
- Head Invigilator, School of Sciences and Technology (Dec 2023 – Apr 2024); Regular Invigilator, Examinations Office (Apr 2023 – Jan 2025).
- “Engineering 2050 for Good” Hackathon (Oct 2023).

Teaching, National Institute of Technology Silchar (2021 – 2023)

- Teaching Assistant: ME-316 Machine Design Laboratory (SolidWorks, ANSYS, FEA); ME-202 Theory of Machines.

Workshop co-organisation

- Co-organiser, M2L (Mechanics and Machine Learning) Workshop, City St George's, July 2024 and July 2025. EPSRC-supported. Speakers: M. Khodaei (Imperial), G. Subhash (Florida), J.-F. Molinari (EPFL), S. Kumar (TU Delft), S. Chakraborty (IIT Delhi), S.A. Ponnusami (City).

Open-Source Software and Online Presence

- **PhAST** – github.com/CEMS-Lab/PhAST – matrix-free, differentiable PyTorch software for phase-field fracture.
- **geometric-transformations** – github.com/allamaprabhuani/geometric-transformations – 2D/3D affine transforms with MATLAB Live Scripts.
- **Personal site** – allamaprabhuani.github.io – research notes, technical tutorials, blog.

Technical Skills

Scientific computing and ML. PyTorch (autograd, CUDA), JAX (familiar), NumPy/SciPy, scikit-learn, pandas, Excel-VBA. Differentiable physics, neural operators (DeepONet, FNO), physics-informed neural networks.

Finite element analysis. Akantu (C++), FEniCSx, COMSOL Multiphysics, MSC Nastran/Patran (SOL-101/103/105/106), Altair Hypermesh/Optistruct, Abaqus, ANSYS, SolidWorks.

HPC and infrastructure. Slurm, Linux clusters, Git, Sphinx, latexmk, reproducible Python packaging (pyproject, editable installs).

Programming. Python, MATLAB, C++ (read), Excel-VBA, $\LaTeX$, Bash.

Reference texts and manuals. E.F. Bruhn (Flight Vehicle Structures), Michael Niu (Airframe Stress Analysis), Boeing Design Manuals, ESDU, Timoshenko (Plates and Shells), Lekhnitskii (Anisotropic Plates), Roark (Stress and Strain).

Referees

Available on request.